Stopping & momentum

Stop safely

1 Thinking & braking

Stopping distance = thinking + braking. Thinking = speed × reaction time. Typical reaction time 0.2–0.9 s.

Tiredness, drugs/alcohol and distractions can delay reaction. Ruler-drop tests: repeat, control setup, consider anticipation.

2 Braking factors

Wet/icy roads, worn tyres/brakes reduce effective braking. Friction transfers kinetic energy to thermal stores.

Same mass/force: double speed → four times braking distance. Thinking only doubles at fixed reaction time. Large deceleration risks overheating/loss of control.

Conserve momentum

3 Momentum is a vector

$p = mv$, kg m/s. Choose a positive direction and use signed velocities.

Closed system with zero resultant external force: total momentum before = after.

4 Collisions & recoil

2 kg at 3 m/s joins stationary 1 kg: 6 kg m/s / 3 kg = **2 m/s.**

Recoil from rest: equal opposite momenta, lighter mass moves faster. Kinetic energy need not be conserved.

Reduce impact forces

5 Force changes momentum

$F = \Delta p/\Delta t = m(v-u)/\Delta t$. Gives average force for the interval.

60 kg at 10 m/s → rest: $|\Delta p| = 600$ kg m/s. Over 0.20 s → 3,000 N; over 0.50 s → 1,200 N.

6 Safety features

Airbags, belts, helmets, mats and soft surfaces increase stopping time for the same momentum change.

Average force is lower. They do not stop momentum changing or destroy energy.